

Homework 5

Showkot Hossain

Part 1: Conceptual Questions

Q1:

No, this regular expression is not safe. The problems are:

1. **ReDoS Vulnerability:** The combination of non-greedy quantifiers (`.*?`) and the character class in `<body[^>]*>` can cause catastrophic backtracking. If input contains many `<body` sequences without closing `>`, the regex engine may take exponential time, causing Denial of Service.
2. **Dot/Newline Issue:** The dot (`.`) doesn't match newlines by default in many engines. Attackers can bypass validation by inserting newlines, or valid multiline HTML may fail to match.
3. **HTML Parsing Limitations:** Regular expressions cannot properly parse HTML (a context-free language). This regex is easily fooled by nested tags, comments (`<!-- </html> -->`), or tag-like strings, making it unsafe for security decisions.

Q2:

To reach `printf("everywhere\n")`, the following conditions must be satisfied:

1. First if `(x > 5)` is true: $x > 5$
2. Second if `(y > 7)` is false (enter else): $y \leq 7$ (i.e., $\neg(y > 7)$)
3. Third if `(x < 20)` is true: $x < 20$

Therefore the path condition is: $(x > 5) \wedge (y \leq 7) \wedge (x < 20)$